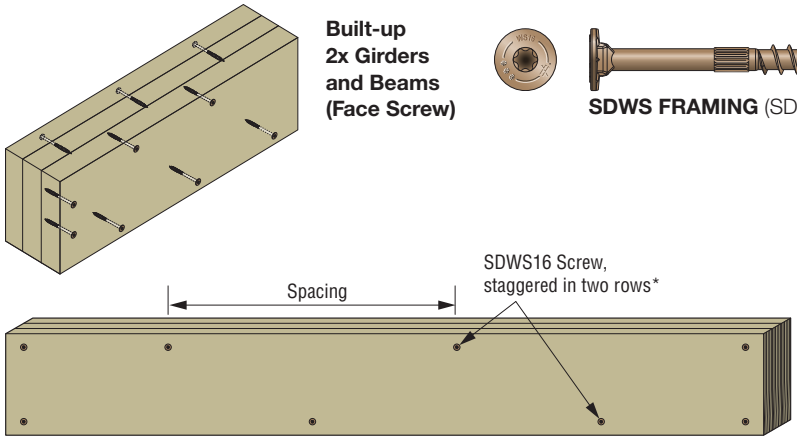
Beam Assemblies and Beam-to-Post Connections

Multi-Ply Beam Assemblies

For multi-ply beam assemblies, the Strong-Tie® SDWS™ Framing screw can be used as an alternative to 10d (0.128" x 3") box nails, when installed in the following configuration using 16" o.c. spacing.



Beam Assembly Detail*

*Fastening pattern shown applies to each ply of the built-up 2x beam.

Building Code Requirements

- Deck beams shall be fastened together with two rows of 10d (0.128" x 3") box nails minimum at 16" o.c. along each edge.

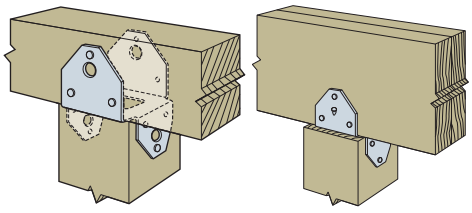
  IRC 2018/2021/2024 Section R507.5

  IBC 2018 Section 2304.10.1
  IBC 2021/2024 Section 2304.10.2

- Deck beams shall be attached to wood posts in accordance with Figures R507.5.1(1) and R507.5.1(2), Figures R507.5.2 (1) and R507.5.2 (2) for IRC 2024 or by other equivalent means capable to resist lateral displacement.

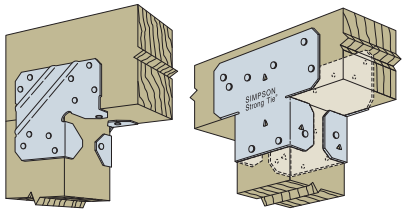
  IRC 2018/2021/2024 Section R507.5.2

Beam-to-Post Connections



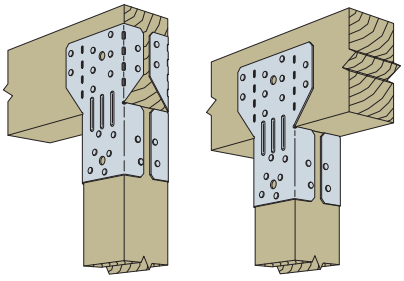
BC/BCS Post Cap

BC: For single-member solid sawn beams. BCS: Connects double 2x's to a 4x post or triple 2x's to a 6x post. ZMAX® coating or stainless steel recommended.



LCE™/AC Post Caps

Two-piece cap should be installed in pairs before or after lumber is in place when the sides of the post and beam are flush. ZMAX coating or stainless steel recommended.



ACHZ™ Post Caps

Two-piece cap should be installed in pairs and may be installed before or after lumber is in place.

At the point where a beam meets a post, it must be properly connected to the post in order to resist gravity, lateral and uplift loads. This pertains to solid sawn beams or those comprised of multiple members, whether they rest on top or are fastened to the side of the post. For one-sided installations with LCE/AC and ACHZ post caps, refer to L-C-ONEPOSTCAP on **strongtie.com**.

F-DECKCODE25 © 2025 Simpson Strong-Tie Company Inc.

Beam-to-Post Connections

Building Code Requirements

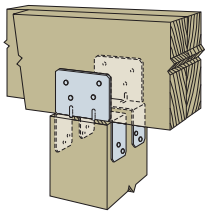
- Deck beams shall be attached to wood posts in accordance with Figures R507.5.1(1) and R507.5.1(2), Figures R507.5.2 (1) and R507.5.2 (2) for IRC 2024 or by other equivalent means capable to resist lateral displacement.

  IRC 2018/2021/2024 Section R507.5.2

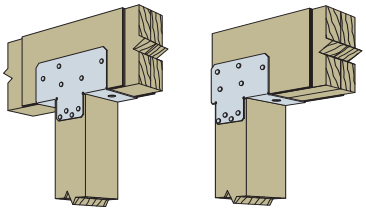
- Where posts and beam or girder construction is used to support floor framing, positive connections shall be provided to ensure against uplift and lateral displacement.

  IRC 2018/2021/2024 Section R502.9
  IBC 2018 Section 2304.10.7
  IBC 2021/2024 Section 2304.10.8

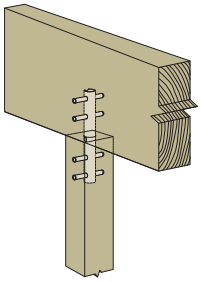
Simpson Strong-Tie® Solutions



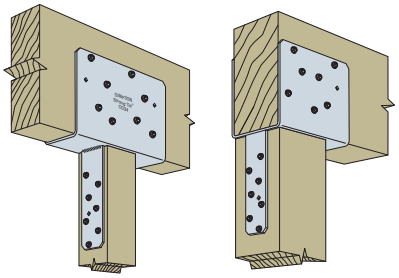
LPCZ Post Cap
Two-piece cap adjusts for beams smaller than post width. Features a ZMAX® coating.



PCZ™/EPCZ™ Post Caps
Connects beams at the top of the post. Features ZMAX coating. Installation without shims, similar.



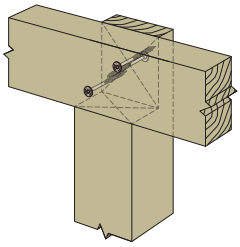
CBTZ® Concealed Beam Tie
Provides clean look for beam-to-post connections. ZMAX coating.



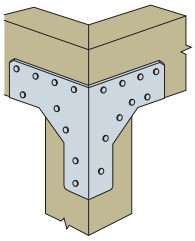
CCQ™/ECCQ™ Beam-to-Post Connections
Provides a strong connection for column-beam combinations. *(HDG or stainless steel)*

The following are alternative connections for select beam-to-post assemblies.

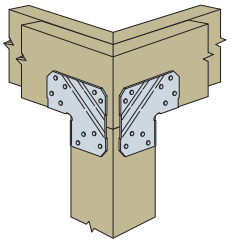
Beam-to-Post Connection



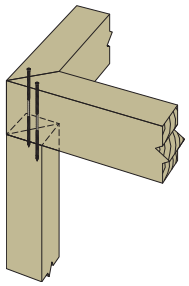
Beam-to-Notched Post, Non-Spliced
Uses Strong-Drive® SDWS™ Timber or SDWS Timber SS Screws



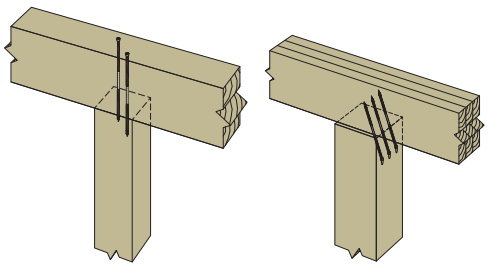
RTC44Z
Provides for a mitered beam-to-post corner. ZMAX coating.



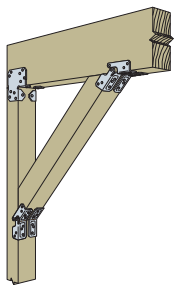
LCE4
Provides for a mitered beam-to-post corner. ZMAX coating or stainless steel.



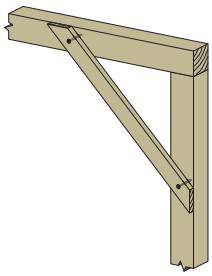
Mitered Beam over 6x Corner Post
Uses Strong Drive SDWH™ Timber-Hex HDG or SWD Double-Threaded™ screw



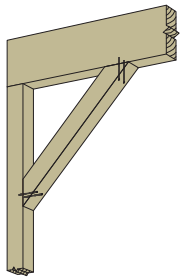
Continuous Beam over 6x Post
Uses Strong Drive SDWH Timber-Hex HDG or SWD Double-Threaded screws



KBS1Z®
Connects diagonal bracing to beams and columns. ZMAX coating. *(If using an MPB, diagonal bracing may not be necessary. Contact a design professional for additional information.)*



Alternative Diagonal Bracing
Uses Strong-Drive SDWS Timber, SDWS Timber SS SDWH Timber-Hex HDG or SWD Double-Threaded screws*



Knee Bracing Uses Strong-Drive SWD Double-Threaded Screws
Connects knee bracing to beams and columns.

*For more information on fastener details to meet DCA6, refer to C-F-2025TECHSUP *Fastener Technical Guide*.